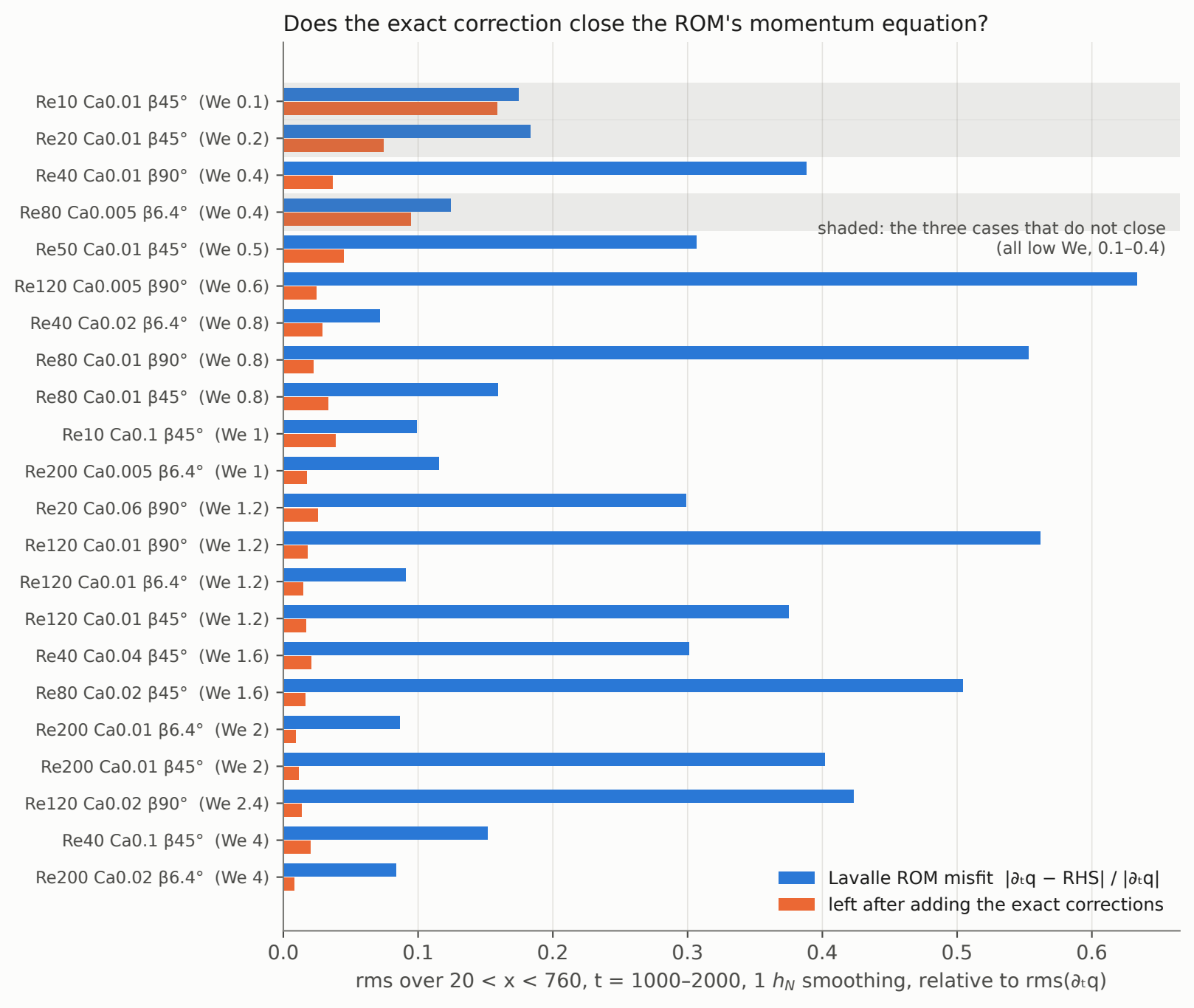

Does the exact correction close the ROM's momentum equation?
Re10 Ca0.01 β45° (We 0.1)
Re20 Ca0.01 β45° (We 0.2)
Re40 Ca0.01 β90° (We 0.4)
Re80 Ca0.005 β6.4° (We 0.4)
Re50 Ca0.01 β45° (We 0.5)
Re120 Ca0.005 β90° (We 0.6)
Re40 Ca0.02 β6.4° (We 0.8)
Re80 Ca0.01 β90° (We 0.8)
Re80 Ca0.01 β45° (We 0.8)
Re10 Ca0.1 β45° (We 1)
Re200 Ca0.005 β6.4° (We 1)
Re20 Ca0.06 β90° (We 1.2)
Re120 Ca0.01 β90° (We 1.2)
Re120 Ca0.01 β6.4° (We 1.2)
Re120 Ca0.01 β45° (We 1.2)
Re40 Ca0.04 β45° (We 1.6)
Re80 Ca0.02 β45° (We 1.6)
Re200 Ca0.01 β6.4° (We 2)
Re200 Ca0.01 β45° (We 2)
Re120 Ca0.02 β90° (We 2.4)
Re40 Ca0.1 β45° (We 4)
Re200 Ca0.02 β6.4° (We 4)
shaded: the three cases that do not close
(all low We, 0.1–0.4)
Lavalle ROM misfit |∂ₜq − RHS| / |∂ₜq|
left after adding the exact corrections
0.0
0.1
0.2
0.3
0.4
0.5
0.6
rms over 20 < x < 760, t = 1000–2000, 1 h_N smoothing, relative to rms(∂ₜq)